

Abil Varkichan Jose

Dallas, TX | [linkedin.com/in/abilvjose](https://www.linkedin.com/in/abilvjose) | github.com/abilvj2 | abilvj2.github.io/Portfolio

Intelligent Automation / AI Engineer with 7+ years designing automated, intelligent and scalable solutions across healthcare, enterprise and consulting domains. Experienced across the full AI development lifecycle — data ingestion through deployment and monitoring — including robotic process automation, Large Language Model applications, Retrieval-Augmented Generation and generative AI agents on Azure and Microsoft Copilot Studio. Strong foundation in traditional machine learning (TensorFlow, PyTorch, Scikit-learn), RESTful APIs, containerised deployment (Docker, Kubernetes) and CI/CD, with HIPAA-aligned healthcare data experience serving 500+ users.

TECHNICAL SKILLS

Intelligent Automation & RPA: UiPath, Robot Framework, Power Automate, Power Apps (Canvas & Model-Driven), Power Virtual Agents, Attended & Unattended Bots, Desktop & Web Automation, Document Processing (Textextract, Document AI), Process Mining (Celonis), End-to-End Test Automation

Generative AI & LLM: Large Language Models, Retrieval-Augmented Generation (RAG), Agentic RAG, Generative AI Agents, LangChain, Pydantic AI, Microsoft Copilot Studio, Agent Orchestration, Prompt Engineering, OpenAI / Azure OpenAI API, Function Calling / Tool Use

ML & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, SHAP, Pandas, NumPy, NLP, Computer Vision, CNNs, Neural Networks

Languages: Python, C#.NET, JavaScript, TypeScript, Java, C/C++, SQL, VBA, HTML/CSS

Cloud & DevOps: Azure (OpenAI, Entra ID, App Service, Fabric), AWS (EC2, S3, Textextract, Athena), Google Cloud (Document AI, Vision API), Docker, Kubernetes, CI/CD, Git/GitHub

APIs & Web: RESTful APIs, FastAPI, Flask, Django, Node.js, React, Angular, Spring Boot, MERN, SDLC

Data & Big Data: Data Pipelines / ETL, Microsoft Fabric, Power BI, Apache Spark, Hadoop, Kafka, Scala, Tableau, Alteryx, pgvector

Databases & Methods: SQL Server, MySQL, PostgreSQL, MongoDB; Agile, Scrum, Kanban (Jira)

PROFESSIONAL EXPERIENCE

Software Engineer — Creative Solutions in HealthCare

Jul 2024 – Present

Fort Worth, Texas

- Engineered and deployed **AI-powered agents** on Microsoft Copilot Studio and Azure OpenAI, applying **retrieval-augmented generation (RAG)** over enterprise healthcare data to automate knowledge workflows for **500+ employees**.
- Designed and maintained **data pipelines on Microsoft Fabric** for ingestion, processing and transformation of large healthcare datasets, delivering real-time analytics and cutting turnaround time by 5 hours.
- Created Model-Driven and Canvas Apps in Power Apps and automated business processes with Power Automate, reducing manual tasks and increasing operational efficiency.
- Built Python and C# services with RESTful API integrations connecting Power Platform applications, SharePoint and SQL Server, monitoring performance and reliability in production.
- Designed and optimised 3+ SQL databases with role-based access control via **Azure Entra ID**, ensuring HIPAA-aligned handling of PHI while improving query performance for high-volume applications.

Graduate Research Assistant — Cross-Platform Developer — Computer Science & Engineering, UT Arlington

Oct 2022 – May 2024

Arlington, Texas

- Built **robotic process automation** with Robot Framework achieving a **70x reduction in human input** for medical form processing, integrating Google Cloud Document AI.
- Created an interactive web portal in **Django** as the end-user platform for the automation, contributing to community projects.
- Applied advanced image and sensory data interpretation in support of community-benefit research.

Automation & Data Visualization Architect (Intern) — HP (Hewlett Packard)

May 2023 – Aug 2023

Houston, Texas

- Spearheaded design and implementation of enterprise **Power BI** dashboards for business pipeline funnel, operational metrics and automation platform performance, delivering real-time visibility to **100+ stakeholders**.
- Engineered intelligent automation for real-time monitoring, **enhancing data-driven decision-making by 70%**; developed ETL pipelines on Azure for seamless multi-source integration.
- Engineered analytics metrics and KPIs using generative AI (GPT, Power Virtual Agents), mining AR, AP, T&E and Payroll data via Webi, Celonis, SQL and Teradata.
- Spearheaded a reporting roadmap and governance process with IT, Finance and data partners, aligning with S/4HANA deployment and stabilisation.

Senior Software Engineer — Infosys Ltd

Oct 2019 – Jun 2022

Mysuru, Karnataka, India

- Led a team developing **AI-enabled automation solutions and bots** for financial-services clients, improving production efficiency by **50%+** in the client business environment.
- Built desktop and web automation integrating Google Drive APIs with end-to-end testing, **reducing processing time from 60 to 5 minutes**; automated Oracle Cloud journal creation and optimised SQL calls for **5x faster** execution.
- Developed full-stack applications in **React.js, Redux and Python**; built computer vision and deep learning backends with Python and OpenCV, deploying on Azure App Service.
- Trained **30+ associates** on Power Platform; served as Scrum Master and Business Analyst, driving Canvas and Model-Driven app development.

SELECTED PROJECTS

- **Prior-Authorization Copilot — Agentic RAG** (2026): Abstention-first decision support for prior-authorization review. Typed and cited output with no *APPROVED* value in the enum, so it cannot issue a decision. Pydantic AI, FastAPI, Postgres + pgvector, XGBoost + SHAP. [GitHub](#)
- **Robot Framework Web Extraction & PDF Filling** (2023): End-to-end insurance claim processing with no human intervention — extracts insurance data from the web, runs calculations, fills clinic software and billing PDFs. [GitHub](#)
- **Face-to-Sketch with CycleGAN** (2023): Unpaired image-to-image translation converting facial photographs into artistic sketches. [GitHub](#)
- **Face Recognition Enabled Browser** (2019): Privacy browser granting access only while the recognised user is present at the webcam, built on OpenCV. [GitHub](#)
- **GCP Traffic Sign Classifier** (2020): CNN in Keras/TensorFlow classifying traffic signs, deployed on Google Cloud Platform. [GitHub](#)

EDUCATION

M.S. Computer Science University of Texas at Arlington, TX	May 2024
B.Tech Computer Science Viswajyothi College of Engineering & Technology APJ Abdul Kalam Technological University, Kerala, India	May 2019

SPEAKING & AWARDS

- **Invited Speaker** — “Power Platform and Copilot Agents”, Arlington Microsoft Fabric User Group (May 2026)
- **Best People’s Choice Intern Award** — business impact across global HP interns (2023)
- **Project Excellence Award** — Microsoft Business Application Practice, Infosys
- **Volunteer** — FIFA World Cup 2026

CERTIFICATIONS

- Microsoft Certified: **Azure AI Fundamentals (AI-900)** — Jun 2022 • Microsoft Certified: **Power Platform Fundamentals (PL-900)** — Dec 2021
- Introduction to Artificial Intelligence with Python — HarvardX CS50, Aug 2020 • Introduction to TensorFlow for AI, ML and Deep Learning — Coursera, May 2020 • Image Classification with CNNs using Keras — Coursera, May 2020